

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Winter Examination-2023 Course: B. Tech. in Mechanical Engineering Branch : Mechanical & Allied Semester :VIII Subject Code & Name: BTMEC801A Fundamental of Automotive System Max Marks: 60 Date:15-01-24 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Explain the working four stroke CI engine	CO1	6
B)	Explain the components of SI Engine with suitable schematic diagram.	CO1	6
C)	For the air standard dual cycle, the compression ratio of 8, cut off ratio 2 and a pressure ratio of 2. Find the air standard dual cycle thermal efficiency.	CO1	6
Q.2	Solve Any Two of the following.		12
A)	A four cylinder engine running at 1200 rpm delivers 20 kW. The average torque when one cylinder was cut is 110 Nm. Find the indicated thermal efficiency if the calorific value of the fuel is 43 MJ/kg and the engine uses of gasoline per kWh.	CO2	6
B)	Explain different phases of Combustion in CI Engine.	CO2	6
C)	Write a short note on Knocking and Detonation.	CO2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain catalytic analyzer with its advantages and disadvantages.	CO3	6
B)	With neat sketch explain Friction Clutch and explain its advantages and disadvantages.	CO4	6
C)	Derive the expression Tractive force available at tyre road interface in the powertrain analysis.	CO4	6
Q.4	Solve Any Two of the following.		12
A)	Explain synchromesh gear transmission system and give its advantages.	CO4	6
B)	With neat sketch explain working of internal expanding brake and enlist its advantages.	CO5	6
C)	With neat sketch explain working of band brake and enlist its	CO5	6

	advantages.		
Q. 5	Solve Any Two of the following.		12
A)	Explain working of Ackerman's Steering system.	CO5	6
B)	Explain signs of poor wheel alignment.	CO6	6
C)	With neat sketch explain working of twin tube shock absorber	CO6	6
	*** End ***		

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